#ManyDaughters: Pre-analysis plan

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H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

I Variables

Dependent variable

equalwork = Equal work (shared responsibility). To measure support for gender equality (attitude), we use the variable **plh0308_v2** (Best if man and woman work the same amount so they can share the responsibility). We used this variable since men tend to work more hours per day than women in terms of *paid* labor. Therefore, more gender equality reflects more support for the statement. Agreement with the statement (Best if man and woman work the same amount so they can share the responsibility) was assessed using a 7-point Likert scale, where higher scores reflect stronger agreement.

Main independent variable

d_daughter_abs = Treated with at least one daughter. We used the variable **pld0026** ("Number of Daughters") and transposed it into a binary variable, with 1 or higher denoting "yes." The answer "does not apply" equals "No." Noteworthy, some people were asked about the number of daughters in one year, but not in others. However, this reduces the sample size to a large extent. Therefore, we apply an absorbing approach, meaning that when person / household is treated, the status is carried forward in subsequent time periods.

Controls

sex = Male (1) and Female (2) – no changes to the original code

bioage = Age of the people – no changes to the original code

bioage_sq = To capture non-linearity in age, we measure bioage²/100

hghinc = Monthly Household Net Income (EUR) – no changes to the original code

pgbilzeit = Amount Of Education Or Training In Years – no changes to the original code

ple0008 = Current Health – no changes to the original code

II Inclusion / Exclusion

Data availability for equalwork and d_daughter_abs determine the analysis. No observations are excluded intentionally.

III Analysis

Since our dataset includes observations from 2018 (no panel data) and our main variable of interest ranges from 1-7, we estimate an OLS model. We use robust standard errors in OLS to account for heteroskedasticity, ensuring that inference remains valid even if the error terms have non-constant variance.

Our approach to data analysis reads:

$$equalwork_i = \boldsymbol{\beta_1} * \boldsymbol{d_daughter_abs_i} + \beta_2 * sex_i + \beta_3 * bioage_i + \beta_4 * \frac{bioage_i{}^2}{100} + \beta_5 * migback_i + \beta_6 * hghinc_i + \beta_7 * mpgbilzeit_i + \beta_8 * ple0008_i + \varepsilon_i$$

IV Interpretation of results

Our results can be found in Table 1:

Table 1: Results of the regression analysis for Hypothesis 1

```
Linear regression                               Number of obs     =        931
                                                F(9, 921)         =       4.64
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0408
                                                Root MSE          =     1.7328

------------------------------------------------------------------------------
                             |               Robust
                   equalwork |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-----------------------------+------------------------------------------------------------------
                d_daughter_abs |  -.0695289   .1152687    -0.60   0.547    -.2957486    .1566909
                               |
                         sex |
                  [2] Female   |    .248459   .1159054     2.14   0.032     .0209897    .4759283
                       bioage |     .01595    .020288     0.79   0.432    -.0238661    .0557661
                     bioage_sq |   .0031281   .0208526     0.15   0.881    -.0377961    .0440522
                               |
                      migback |
  [2] direct migration background  |   .1499442   .1774993     0.84   0.398    -.1984058    .4982943
[3] indirect migration background  |   .0927689   .3045039     0.30   0.761    -.5048331     .690371
                               |
                        hghinc |   -.000053   .0000329    -1.61   0.107    -.0001176    .0000115
                     pgbilzeit |  -.0102822   .0228923    -0.45   0.653    -.0552094     .034645
                       ple0008 |   .0056072   .0650751     0.09   0.931    -.1221055    .1333198
                         _cons |   4.864915   .5809172     8.37   0.000      3.72484     6.00499
------------------------------------------------------------------------------
```

The null hypothesis of no effect of having at least one daughter on attitudes regarding equal work (shared responsibility) cannot be rejected (P = 0.547). Based on the p-value (threshold for statistical significance 0.05 / 4 due to multiple testing), the **test result is not in line with the hypothesis**. In

other words, our regression analysis does not indicate that having daughters leads to more support of "shared responsibility."

The coefficient is negative, and with a larger sample size, the p-value would decrease by definition. A sample with more observations may find preference in favor of sex-based division of labor.

H2. Having daughters leads to more left-leaning political preferences.

I Variables

Dependent variable

leftlean = Political attitude (left). To measure political attitude, we use the variable **plh0004** ("Political attitude"). Since higher values shall represent more left-leaning political preferences, we reversed the scale (10 - plh0004). Overall, using leftlean seems to be straightforward to analyze H2.

Main independent variable

d_daughter_abs = Treated with at least one daughter. We used the variable **pld0026** ("Number of Daughters") and transposed it into a binary variable, with 1 or higher denoting "yes." The answer "does not apply" equals "No." Noteworthy, some people were asked about the number of daughters in one year, but not in others. However, this reduces the sample size to a large extent. Therefore, we apply an absorbing approach, meaning that when person / household is treated, the status is carried forward in subsequent time periods.

Controls

sex = Male (1) and Female (2) – no changes to the original code

bioage = Age of the people – no changes to the original code

bioage_sq = To capture non-linearity in age, we measure bioage²/100

hghinc = Monthly Household Net Income (EUR) – no changes to the original code

pgbilzeit = Amount Of Education Or Training In Years – no changes to the original code

ple0008 = Current Health – no changes to the original code

II Inclusion / Exclusion

Data availability for leftlean and d_daughter_abs determines the analysis. No observations are excluded intentionally.

III Analysis

Since our dataset includes multiple time periods of observations, we employ a panel data model. One decision to be made was whether to use a RE or FE specification. A Hausman Test rejects the null hypothesis that the Random Effects (RE) model is consistent and efficient ($p = 0.0251$). Therefore, we aimed to use a Fixed Effects panel data model. However, the F-test indicates that we cannot learn anything from this model ($P = 0.8169$). As a consequence, we decided to use the RE model – losing some efficiency, but having an interpretable model. We apply robust standard

errors. To be more precise, the standard errors are clustered at the individual level to account for the potential correlation of error terms within each individual over time.

Our approach to data analysis reads:

$$leftlean_{it} = \boldsymbol{\beta_1} * \boldsymbol{d_daughter_abs_{it}} + \beta_2 * syear_t + \beta_3 * sex_i + \beta_4 * bioage_{it} + \beta_5 * \frac{bioage_{it}^{\ 2}}{100} + \beta_6 * migback_{it} + \beta_7 * hghinc_{it} + \beta_8 * mpgbilzeit_{it} + \beta_9 * ple0008_{it} + \alpha_i + \varepsilon_{it}$$

IV Interpretation of results

Our results can be found in Table 2.

Table 2: Results of the regression analysis for Hypothesis 2

```
Random-effects GLS regression                   Number of obs     =      3,107
Group variable: pid                             Number of groups  =      1,750

R-sq:                                           Obs per group:
     within  = 0.0005                                         min =          1
     between = 0.0341                                         avg =        1.8
     overall = 0.0227                                         max =          4

                                                Wald chi2(12)     =      55.27
corr(u_i, X)   = 0 (assumed)                    Prob > chi2       =     0.0000

                                                      (Std. Err. adjusted for 1,750 clusters in pid)
------------------------------------------------------------------------------------------------------
                             |               Robust
                    leftlean |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-----------------------------+------------------------------------------------------------------------
                d_daughter_abs |   .0363987    .065563     0.56   0.579    -.0921024    .1648997
                             |
                       syear |
                        2009 |   -.011004   .0714115    -0.15   0.878    -.1509679    .1289598
                        2014 |   .1672911   .0759902     2.20   0.028     .0183531    .3162291
                        2019 |   .1586362   .0779729     2.03   0.042     .0058121    .3114604
                             |
                         sex |
                  [2] Female |   .2109936   .0784418     2.69   0.007     .0572506    .3647367
                      bioage |   .0128912   .0115468     1.12   0.264      -.00974    .0355225
                    bioage_sq |  -.0176062   .0117591    -1.50   0.134    -.0406536    .0054411
                             |
                     migback |
  [2] direct migration background |   .2713138   .1169921     2.32   0.020     .0420135    .5006141
[3] indirect migration background |   .3303915   .2102271     1.57   0.116     -.081646     .742429
                             |
                      hghinc |  -.0000366   .0000173    -2.11   0.034    -.0000706   -2.68e-06
                   pgbilzeit |   .0522225   .0141573     3.69   0.000     .0244746    .0799704
                     ple0008 |  -.0180937   .0335775    -0.54   0.590    -.0839044    .0477171
                       _cons |   4.379468   .3369955    13.00   0.000     3.718969    5.039967
-----------------------------+------------------------------------------------------------------------
                     sigma_u |  1.2102471
                     sigma_e |  1.1280647
                         rho |  .53510267   (fraction of variance due to u_i)
------------------------------------------------------------------------------------------------------
```

The null hypothesis of no effect of having at least one daughter on left-leaning preferences cannot be rejected (P = 0.579). Based on the p-value (threshold for statistical significance 0.05 / 4 due to

multiple testing), the **test result is not in line with the hypothesis**. In other words, our regression analysis does not indicate that having daughters leads to more left-leaning political preferences.

However, this could be due to the low sample size. The coefficient is slightly positive, and with a larger sample size, the p-value would decrease by definition.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

I Variables

Dependent variable

man_hours = Hours Weekday Housework (Man). To measure gender equality in the household (behavior/outcome), we use the variable **pli0043_h** (Hours Weekday Housework). The vast majority of empirical studies show that women (still) tend to do the housework. More gender equality would mean that men do more housework. Therefore, we analyze weekday housework in hours done by man (in the following referred to as man_hours).

Main independent variable

d_daughter_abs = Treated with at least one daughter. We used the variable **pld0026** ("Number of Daughters") and transposed it into a binary variable, with 1 or higher denoting "yes." The answer "does not apply" equals "No." Noteworthy, some people were asked about the number of daughters in one year, but not in others. However, this reduces the sample size to a large extent. Therefore, we apply an absorbing approach, meaning that when person / household is treated, the status is carried forward in subsequent time periods.

Controls

sex = Male (1) and Female (2) – no changes to the original code

bioage = Age of the people – no changes to the original code

bioage_sq = To capture non-linearity in age, we measure $\text{bioage}^2/100$

hghinc = Monthly Household Net Income (EUR) – no changes to the original code

pgbilzeit = Amount Of Education Or Training In Years – no changes to the original code

ple0008 = Current Health – no changes to the original code

II Inclusion / Exclusion

Data availability for man_hours and d_daughter_abs determines the analysis. No observations are excluded intentionally.

III Analysis

Since our dataset includes multiple time periods of observations, we employ a panel data model. One decision to be made was whether to use a RE or FE specification. A Hausman Test rejects the null hypothesis that the Random Effects (RE) model is consistent and efficient ($p = 0.0034$). Therefore, we estimate a Fixed Effects panel data model. We apply robust standard errors. To be

more precise, the standard errors are clustered at the individual level to account for the potential correlation of error terms within each individual over time.

Our approach to data analysis reads:

$$man_hours_{it} = \boldsymbol{\beta_1} * \boldsymbol{d_daughter_abs_{it}} + \beta_2 * syear_t + \beta_3 * sex_i + \beta_4 * bioage_{it} + \beta_5 * \frac{bioage_{it}^{\ 2}}{100} + \beta_6 * migback_{it} + \beta_7 * hghinc_{it} + \beta_8 * mpgbilzeit_{it} + \beta_9 * ple0008_{it} + \alpha_i + \varepsilon_{it}$$

IV Interpretation of results

Our results can be found in Table 3.

Table 3: Results of the regression analysis for Hypothesis 3

```
Fixed-effects (within) regression               Number of obs     =     18,650
Group variable: pid                             Number of groups  =      3,099

R-sq:                                           Obs per group:
     within  = 0.0091                                         min =          1
     between = 0.0008                                         avg =        6.0
     overall = 0.0013                                         max =         21

                                                F(24,3098)        =       5.79
corr(u_i, Xb)  = -0.1510                        Prob > F          =     0.0000

                                                      (Std. Err. adjusted for 3,099 clusters in pid)
------------------------------------------------------------------------------------------------------
                                    |               Robust
                         man_hours |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
------------------------------------+-----------------------------------------------------------------
                       d_daughter_abs |  -.0314197   .0326902    -0.96   0.337    -.0955163     .032677
                                    |
                                syear |
                                 2001 |   .0260283   .0452186     0.58   0.565    -.0626332    .1146898
                                 2002 |    .050594    .042711     1.18   0.236    -.0331508    .1343388
                                 2003 |   .0273307   .0466415     0.59   0.558    -.0641208    .1187821
                                 2004 |   .1009869   .0480588     2.10   0.036     .0067565    .1952173
                                 2005 |   .0305859   .0494627     0.62   0.536     -.066397    .1275688
                                 2006 |   .0510531   .0540817     0.94   0.345    -.0549865    .1570928
                                 2007 |   .0284944   .0564606     0.50   0.614    -.0822097    .1391985
                                 2008 |  -.0051614   .0542794    -0.10   0.924    -.1115886    .1012658
                                 2009 |  -.0254645   .0580039    -0.44   0.661    -.1391945    .0882654
                                 2010 |   .0407308   .0556793     0.73   0.465    -.0684413    .1499029
                                 2011 |  -.0375563   .0595276    -0.63   0.528    -.1542739    .0791612
                                 2012 |   .0646962    .061388     1.05   0.292     -.055669    .1850615
                                 2013 |  -.0175583   .0621665    -0.28   0.778    -.1394501    .1043335
                                 2014 |   .0079024   .0656098     0.12   0.904    -.1207407    .1365455
                                 2015 |  -.0342683   .0646579    -0.53   0.596     -.161045    .0925083
                                 2016 |  -.0221856   .0666964    -0.33   0.739    -.1529592    .1085881
                                 2017 |   .1451769   .0680748     2.13   0.033     .0117006    .2786533
                                 2018 |    .007778   .0692261     0.11   0.911    -.1279558    .1435118
                                 2019 |   .1125488   .0692204     1.63   0.104    -.0231737    .2482712
                                 2020 |   .2250316   .0715732     3.14   0.002      .084696    .3653673
                                    |
                               bioage |          0  (omitted)
                            bioage_sq |          0  (omitted)
                                    |
                              migback |
  [2] direct migration background  |          0  (omitted)
[3] indirect migration background  |          0  (omitted)
                                    |
                              hghinc |    .000015   8.89e-06     1.69   0.091    -2.41e-06    .0000324
                            pgbilzeit |  -.0455016   .0360333    -1.26   0.207    -.1161532      .02515
                              ple0008 |   .0165724   .0106557     1.56   0.120    -.0043205    .0374654
                                _cons |   1.210825   .4332577     2.79   0.005     .3613231    2.060326
------------------------------------+-----------------------------------------------------------------
                              sigma_u |  .82284621
                              sigma_e |    .742288
                                  rho |  .55133449   (fraction of variance due to u_i)
------------------------------------------------------------------------------------------------------
```

The null hypothesis of no effect of having at least one daughter on weekly housework done by men cannot be rejected (P = 0.337). Based on the p-value (threshold for statistical significance 0.05 / 4 due to multiple testing), the **test result is not in line with the hypothesis**. In other words, our regression analysis does not indicate that having daughters leads to more housework done by men.

In contrast, the coefficient is slightly negative, and with a larger sample size, the p-value would decrease by definition.

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H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

I Variables

Dependent variable

equalwork = Equal work (shared responsibility). To measure support for gender equality (attitude), we use the variable **plh0308_v2** (Best if man and woman work the same amount so they can share the responsibility). We used this variable since men tend to work more hours per day than women in terms of *paid* labor. Therefore, more gender equality reflects more support with the statement. Agreement with the statement (Best if man and woman work the same amount so they can share the responsibility) was assessed using a 7-point Likert scale, where higher scores reflect stronger agreement.

Main independent variable

d_daughter_abs = Treated with at least one daughter. We used the variable **pld0026** ("Number of Daughters") and transposed it into a binary variable, with 1 or higher denoting "yes." The answer "does not apply" equals "No." Noteworthy, some people were asked about the number of daughters in one year, but not in others. However, this reduces the sample size to a large extent. Therefore, we apply an absorbing approach, meaning that when person / household is treated, the status is carried forward in subsequent time periods.

The main difference from H1 is considering men compared to women. Therefore, the main independent variable becomes an interaction of **d_daughter_abs** and **sex.**

Controls

d_daughter_abs = see above, since the interaction is our main variable of interest, d_daughter_abs is added as a control

sex = Male (1) and Female (0) – We recoded male and female, so that female becomes the control.

bioage = Age of the people – no changes to the original code

bioage_sq = To capture non-linearity in age, we measure bioage²/100

hghinc = Monthly Household Net Income (EUR) – no changes to the original code

pgbilzeit = Amount Of Education Or Training In Years – no changes to the original code

ple0008 = Current Health – no changes to the original code

II Inclusion / Exclusion

Data availability for equalwork and d_daughter_abs determine the analysis. No observations are excluded intentionally.

III Analysis

Since our dataset includes observations from 2018 (no panel data) and our main variable of interest ranges from 1-7, we estimate an OLS model. We use robust standard errors in OLS to account for heteroskedasticity, ensuring that inference remains valid even if the error terms have non-constant variance.

Our approach to data analysis reads:

$$equalwork_i = \beta_1 * d_daughter_abs_i + \beta_2 * sex_i + \boldsymbol{\beta_3} * \boldsymbol{d_{daughter_{abs\,i}}} * \boldsymbol{sex_i} + \beta_4 * bioage_i + \beta_5 * \frac{bioage_i{}^2}{100} + \beta_6 * migback_i + \beta_7 * hghinc_i + \beta_8 * mpgbilzeit_i + \beta_9 * ple0008_i + \varepsilon_i$$

IV Interpretation of results

Our results can be found in Table 4.

Table 4: Results of the regression analysis for Hypothesis 4

```
.                reg equalwork i.d_daughter_abs##i.sex bioage bioage_sq i.migback hghinc pgbilzeit ple0008, robust

Linear regression                               Number of obs     =        931
                                                F(10, 920)        =       4.59
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0423
                                                Root MSE          =     1.7324

------------------------------------------------------------------------------------------------
                               |               Robust
                     equalwork |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------------------------+----------------------------------------------------------------
               1.d_daughter_abs |  -.1861639   .1508379    -1.23   0.217    -.4821902    .1098624
                                |
                            sex |
                    [1] Male    |  -.4040953    .175839    -2.30   0.022    -.7491874   -.0590031
                                |
            d_daughter_abs#sex  |
                  1#[1] Male    |   .2765744   .2365324     1.17   0.243    -.1876314    .7407801
                                |
                        bioage  |   .0156313   .0202187     0.77   0.440    -.0240489    .0553115
                     bioage_sq  |   .0032542   .0207827     0.16   0.876    -.0375328    .0440412
                                |
                        migback |
  [2] direct migration background  |   .1499208   .1781102     0.84   0.400    -.1996287    .4994702
[3] indirect migration background  |   .0883652   .3020292     0.29   0.770    -.5043808    .6811113
                                |
                        hghinc  |  -.0000527   .0000325    -1.62   0.105    -.0001165    .0000111
                     pgbilzeit  |  -.0121983   .0229569    -0.53   0.595    -.0572524    .0328557
                        ple0008 |   .0043055    .064939     0.07   0.947    -.1231402    .1317513
                          _cons |   5.219942   .5767322     9.05   0.000     4.088078    6.351805
------------------------------------------------------------------------------------------------
```

The null hypothesis of no effect of having at least one daughter on attitudes regarding equal work (shared responsibility) for men compared to women cannot be rejected (P = 0.022). Based on the p-value (threshold for statistical significance 0.05 / 4 due to multiple testing), the **test result is not**

in line with the hypothesis. In other words, our regression analysis does not indicate that having daughters leads to more support of "shared responsibility" for men compared to women.

However, without considering multiple testing, the interaction is statistically significant at the 5% level and has a negative sign. In other words, with a larger sample size, the null hypothesis of "Best if man and woman work the same amount so they can share the responsibility" is likely to be rejected. Similar to H1, but to a greater extent, this would indicate a preference in favor of sex-based division of labor.